

# Sangjun Yum

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## EDUCATION

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**New York University**, New York, NY

*B.A. in Computer Science, Minor in Mathematics* — January 2018

## EXPERIENCE

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### FalconX – Senior Software Engineer

*New York, NY · January 2024 – April 2026*

Designed and shipped institutional OTC spot trading infrastructure for the largest digital asset prime brokerage serving institutional clients in the U.S.

- Built and launched a production Request-for-Quote (RFQ) workflow integrated with FalconX's spot execution engine. Soft launch August 2024, full launch October 2024. Cumulative volume of \$11B+ since September 2024, with a peak of \$1.27M in monthly revenue (December 2024) across 338 institutional clients and 350+ supported tokens. Average response time: 13ms.
- Designed and delivered FalconX's integration with DRW Cumberland (live May 2025), routing ~\$300M in monthly volume across 51 token markets at a 98% fill rate. Enabled trading in specific tokens (e.g., MOODENG) only available through this integration.
- Architected and launched the Wincent integration (live December 2025), routing ~\$357M in monthly volume across 27 token markets at a 97% fill rate. Together with Cumberland, the integrations handle ~\$657M in combined monthly volume across 78 markets.
- Implemented a statistical pricing model to dynamically adjust trade padding in real time, and ran client-level A/B testing to optimize pricing parameters, PnL, and trading volume.
- Led low-latency execution improvements including FIX-based connectivity adoption, network architecture redesign, regional server relocation to reduce geographic latency, private connectivity to priority liquidity providers, and CPU-level optimizations using x86 assembly on hot paths.
- Managed U.S. trading-hour operational support for institutional clients.

### Grubhub – Senior Software Engineer

*New York, NY · June 2022 – November 2023*

Developed subscription and partnership features supporting long-term strategic revenue.

- Led a team of three to optimize large-scale cashback data performance in Cassandra, reducing monthly read timeouts by 70%.
- Designed and built APIs and user onboarding flows for credit card partnership sign-ups.
- Implemented nightly tasks to evict expired cashback records, improving the accuracy of revenue reporting.
- Contributed to the migration of Lyft Pink partnership users from lifetime to annual membership in support of strategic revenue optimization.

### Goldman Sachs – Software Engineer (Associate)

*New York, NY · June 2021 – June 2022*

Built trade booking and order management systems for the Consumer and Private Wealth Management division's front office, focused on structured notes and derivatives.

- Digitized the trade lifecycle for structured notes and derivative products, including issuance, booking, secondary trading, compliance, and regulatory reporting.
- Designed and implemented automated revenue reporting and structured note issuance reporting routines, replacing manual end-of-day back-office workflows.
- Implemented trade/order FIDLEG validation procedures for newly tradable assets, and contributed to MiFID II compliance systems for European and Swiss clients.
- Contributed to the firm-wide initiative to reduce SecDB and Slang programming language footprint and modernize legacy trading infrastructure for improved data retention, retrieval performance, and trading-desk efficiency.

### SIMON Markets – Platform / Backend Engineer

*New York, NY · May 2019 – June 2021*

Joined as the 7th employee at SIMON Markets, a Goldman Sachs spin-off building the first end-to-end digital issuance and trading platform for structured financial products in the United States. (Now part of iCapital.)

- Refactored a monolithic legacy codebase into a microservices architecture to enable cloud deployment.

- Led migration of SIMON from Goldman Sachs' internal infrastructure to AWS ECS, enabling external client access for the first time.
- Integrated OpenAPI specifications and the Kong API Gateway for service exposure, security, and scalability.
- Designed and built the distributed portfolio and holdings management systems used by financial advisors at major U.S. broker-dealer networks (including Raymond James) to store, retrieve, and manage structured product portfolios.
- Integrated major structured product issuers — Goldman Sachs, Wells Fargo, and HSBC — onto the platform for digital issuance and lifecycle management.
- Built ETL pipelines and educational content delivery systems used by financial advisors learning to evaluate structured products.

## **Cello Capital – Technology Analyst**

*New York, NY · March 2018 – March 2019*

Developed fixed-income analytics and pricing software used internally by Agency MBS traders at an asset management firm specializing in mortgage-backed securities.

- Implemented the backend of a real-time security pricing service used by portfolio managers during live trading sessions.
- Integrated pricing pipelines with external market data providers, including Bloomberg and Citi Yieldbook.
- Built P&L computation pipelines and overnight spread-calculation jobs.
- Supported large-scale quantitative risk computation, including Monte Carlo simulations on high-performance computing infrastructure.

## **PUBLICATIONS**

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### **3 Minute Deep Learning: PyTorch**

Kim K., Yum S. — Hanbit Media, 2019. ISBN: 979-1-1622-4227-8.

Korean-language technical reference on PyTorch. Bestseller in Korea's IT category. Held in 45+ institutional libraries across Korea, including the National Library of Korea, the National Assembly Library, KAIST, Seoul National University, Korea University, Yonsei University, and Sungkyunkwan University.

## **ACTIVITIES**

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### **HackUSF 2026 — General Track Judge** · March 28–29, 2026, Tampa, FL

Invited as a General Track Judge for the University of South Florida's annual hackathon (a Major League Hacking Member Event). Evaluated 85 project submissions from ~290 participants across software engineering and financial technology applications.

## **SKILLS**

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**Languages & Frameworks:** Scala, Python, Java, Go, TypeScript, C#, C, C++, x86 Assembly

**Cloud & Deployment:** AWS (ECS, ECR, S3, RDS), Pulumi, Terraform, Spinnaker, Docker, Kubernetes, ArgoCD

**Messaging & RPC:** Kafka, ActiveMQ, Aeron, gRPC, FIX protocol

**Databases:** PostgreSQL, MongoDB, Cassandra, Redis, Spark SQL; Flyway DB migrations

**API Management:** Kong HQ, OpenAPI, Swagger

**Machine Learning:** PyTorch, TensorFlow, SciPy

**Financial Domain:** Structured products, fixed income, mortgage-backed securities, derivatives, institutional OTC crypto, FX, options pricing (Monte Carlo, Black-Scholes), market microstructure, regulatory compliance (MiFID II, FIDLEG)